

Advanced Concepts(Abstract & Inner Classes)

1. Shape Hierarchy

Task:

Create an abstract class `Shape` with the following:

- An abstract method `double area()`
- An abstract method `double perimeter()`

Then create two concrete subclasses:

- `Circle` (with a `radius`)
- `Rectangle` (with `length` and `width`)

Instructions:

- Override the `area()` and `perimeter()` methods in both subclasses.
 - Write a test class that creates a list of shapes and prints their area and perimeter.
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2. Employee Payment System

Task:

Create an abstract class `Employee` with:

- Fields: `name`, `id`
- An abstract method `double calculateSalary()`

Create two subclasses:

- `HourlyEmployee` (with `hoursWorked`, `hourlyRate`)
- `SalariedEmployee` (with `monthlySalary`)

Instructions:

- Implement `calculateSalary()` for both subclasses.
- Create objects of each type and calculate their salary.

3. Vehicle Example

Task:

Create an abstract class `Vehicle` with:

- Method `startEngine()`
- Method `stopEngine()`
- Abstract method `drive()`

Create two subclasses: `Car` and `Bike`, each with its own `drive()` implementation.

Inner Class – Coding Questions

4. University and Department

Task:

Create a class `University` with an inner class `Department`.

Requirements:

- `University` has a `name`
- `Department` has a `deptName` and method `printDetails()` that prints both university and department names.

Instructions:

- Instantiate the inner class from a `University` object.
- Demonstrate access to outer class members from the inner class.

5. Outer and Static Inner Class

Task:

Create a class `Outer` with:

- A static nested class `StaticInner` that prints a message
- A non-static inner class `Inner` that accesses a private field of the outer class

Instructions:

- Create objects of both inner classes.
- Show differences in how they are instantiated and used.

6. Anonymous Inner Class Example

Task:

Create an interface `Greeting` with a method `sayHello()`.

Create a main class that uses an **anonymous inner class** to implement `Greeting`.

Instructions:

- Instantiate the anonymous inner class inside a method and call `sayHello()`.